Loan Final Outcome Prediction: Data Cleaning, Landmark Features, and Baseline Models

CO-KTU Workshop Credit Platform Analysis

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1 Objective

The modelling objective is to predict the final lifecycle outcome of an issued loan that is still active at a given observation point. The unit of analysis is a loan, identified by `credit_id`, not a borrower. A borrower may therefore contribute multiple loan-level observations.

The current version uses a landmark modelling approach. Separate prediction datasets are constructed for the 30-day, 60-day, and 90-day landmarks. For a landmark $t \in \{30, 60, 90\}$, only information observed from loan origination through day t is used:

$$X_i(t) = \{\text{static features}_i,\ \text{repayment and schedule summaries up to day } t\}.$$

The target is the final terminal outcome after the loan lifecycle has ended.

2 Final Outcome Labels

Only terminal loans are used for supervised training. Loans that had not reached a terminal state by the data extraction date are treated as right-censored and excluded from the first-pass classification training dataset.

The five mutually exclusive terminal labels are assigned by the following priority rule:

1. **Sold**: `sold` has a valid date.
2. **Written Off**: `writeoff` or `writeoff_a` is positive, excluding loans already labelled as sold.
3. **Clean Paid Off**: `paid` has a valid date, final debt is zero, `max_delay = 0`, and `late_fee = 0`.
4. **Paid Off After Delinquency**: `paid` has a valid date and final debt is zero, but delay or fee signals exist.
5. **Other Terminal / Special Closure**: terminal or special closure signals exist but do not fit the four main labels.

This priority rule forces each terminal loan into exactly one final outcome class.

3 Cleaned Modelling Dataset

The cleaned loan-level dataset is saved as:

`data/modeling/loan_modeling_dataset.csv`

It contains 18,851 terminal loan rows and 116 columns. Table 1 shows the label distribution. The rare classes, especially *Written Off* and *Other Terminal / Special Closure*, are very small and should be interpreted cautiously.

Table 1: Final outcome label distribution in the cleaned dataset.

Final outcome	Loans
Clean Paid Off	10,318
Paid Off After Delinquency	8,097
Sold	398
Written Off	28
Other Terminal / Special Closure	10

4 Landmark Eligibility

A loan is eligible for a landmark model only if its terminal date is later than the landmark day. For example, a loan that ends after 45 days is eligible for the 30-day model, but not for the 60-day or 90-day model.

Table 2: Eligible terminal loans by landmark.

Landmark	Eligible terminal loans
30 days	12,895
60 days	9,057
90 days	7,254

5 Feature Construction

The feature set combines conservative static variables and landmark-window summary variables. Variables that are only known after loan termination are excluded from model training, including `terminal_date`, `loan_duration_days`, `final_debt`, `final_max_delay`, and `final_late_fee`.

5.1 Static Features

Static features include loan contract variables and conservative borrower variables:

- loan amount, period, charge, interest, administrative fee, APR, product and schedule type;
- contract ratios such as charge-to-principal and amount per contract day;

- borrower age, gender, income, dependents, work status, home status, region, and customer category;
- credit-level SAS score when available.

Some customer-master aggregate fields, such as `credits_paid`, `last_payment_date`, and `all_paid`, are intentionally excluded because they are likely measured at extraction time and may contain future information relative to the landmark.

5.2 Window Features

For each landmark $t \in \{30, 60, 90\}$, the script constructs features using only events dated on or before $\texttt{loan\_created} + t$ days.

Repayment features come from `payments_credits_202606081635.csv` joined to `payment_202606081636.csv` using:

$$\texttt{payments\_credits.payment\_id} = \texttt{payment.id}.$$

The loan-level join is:

$$\texttt{payments\_credits.credit\_id} = \texttt{export\_credits.id}.$$

For each window, the following actual repayment features are calculated:

- payment count and total paid amount;
- principal paid, interest paid, administrative fee paid, late fee paid, penalty paid, and charge paid;
- extension-related payment count and total;
- days to first payment and days since last payment at the landmark.

Scheduled repayment features come from `credits_schedules_all.csv`:

- scheduled due installments;
- scheduled due total, principal, interest, and administrative fee.

The model also uses derived repayment-performance features:

$$\texttt{payment\_gap}_{i,t} = \texttt{scheduled\_due\_total}_{i,t} - \texttt{payment\_total}_{i,t},$$

$$\texttt{principal\_recovery\_rate}_{i,t} = \frac{\texttt{principal\_paid}_{i,t}}{\texttt{loan\_amount}_i},$$

$$\texttt{paid\_to\_due\_ratio}_{i,t} = \frac{\texttt{payment\_total}_{i,t}}{\texttt{scheduled\_due\_total}_{i,t}}.$$

Extension request features come from `export_credit_ext_requests.csv`: request count, completed extension count, extension price, and extension period sum.

6 Training Procedure

For each landmark, the eligible terminal loans are split into train, validation, and test sets using stratified random sampling with an approximate 60/20/20 split within each final-outcome class.

Two baseline models are trained:

1. **Multinomial logistic regression**: a linear, interpretable baseline.

2. **Random forest**: a tree-based nonlinear baseline.

The local runtime did not include LightGBM, XGBoost, or CatBoost. Therefore, random forest is used as the first tree-based baseline. The same cleaned dataset can later be used to train a gradient boosting model.

7 Model Results

Table 3 summarizes validation and test performance. Accuracy, macro F1, weighted F1, and log loss are reported. Macro F1 is important because the final outcome classes are highly imbalanced.

Table 3: Model performance by landmark.

Landmark	Model	Accuracy	Macro F1	Weighted F1	Log loss
30d	Logistic regression	0.7143	0.5650	0.7261	0.7043
30d	Random forest	0.7708	0.6739	0.7702	0.4994
60d	Logistic regression	0.7475	0.5534	0.7561	0.6150
60d	Random forest	0.7959	0.6967	0.7926	0.4699
90d	Logistic regression	0.7591	0.6221	0.7696	0.7403
90d	Random forest	0.7927	0.5717	0.7866	0.4479

The random forest consistently outperforms multinomial logistic regression in accuracy, weighted F1, and log loss. The 60-day random forest gives the strongest overall test result, with accuracy of 0.7959 and macro F1 of 0.6967.

The rare terminal classes remain difficult to predict. In particular, *Written Off* and *Other Terminal / Special Closure* have very few observations. Their precision and recall estimates are therefore unstable and should not be over-interpreted.

8 SHAP-Based Model Interpretation

SHAP analysis is performed for the 60-day random forest, because this model has the strongest overall test performance. A Monte Carlo Shapley approximation is used on a stratified test sample. To keep computation tractable, the analysis is restricted to the top 20 random-forest importance features. For each sampled loan, the algorithm starts from a baseline observation and repeatedly adds features in random order, measuring the marginal change in each final-outcome probability.

The SHAP values are therefore class-specific probability contributions. Larger mean absolute SHAP values indicate stronger influence on the probability of a given final outcome. Table 4 through Table 7 show the top SHAP features for the main outcome classes.

The interpretation is consistent with the feature-importance results. Early repayment behaviour dominates: late-fee payments, repayment gap, principal recovery rate, first-payment timing, total

Table 4: Top SHAP features for *Clean Paid Off*, 60-day random forest.

Feature	Mean abs. SHAP	Mean SHAP
`w60d_principal_recovery_rate`	0.0495	-0.0224
`w60d_days_to_first_payment`	0.0478	-0.0019
`w60d_late_fee_paid`	0.0471	-0.0471
`customer_category_id`	0.0438	-0.0435
`w60d_total_paid_to_due_ratio`	0.0426	-0.0426
`w60d_payment_gap`	0.0408	-0.0373

Table 5: Top SHAP features for *Paid Off After Delinquency*, 60-day random forest.

Feature	Mean abs. SHAP	Mean SHAP
`customer_category_id`	0.0595	-0.0498
`w60d_late_fee_paid`	0.0499	0.0499
`w60d_days_to_first_payment`	0.0471	0.0049
`w60d_principal_recovery_rate`	0.0408	0.0179
`w60d_total_paid_to_due_ratio`	0.0346	0.0346
`w60d_payment_gap`	0.0343	0.0340

paid-to-due ratio, and payment count are among the most influential predictors. Contract terms such as amount, charge, interest, and administrative fee also matter, but the 60-day repayment behaviour is more informative than static loan terms alone.

9 Limitations and Next Steps

This is a first-pass baseline modelling exercise. The main limitations are:

- severe class imbalance for *Written Off* and *Other Terminal / Special Closure*;
- exclusion of active right-censored loans from the supervised classification training set;
- use of random forest rather than a tuned gradient boosting model due to local package availability;
- approximate SHAP analysis based on a sampled test set and top-ranked features.

Recommended next steps are:

1. train a gradient boosting model such as LightGBM, XGBoost, or CatBoost once available;
2. consider merging very rare terminal classes or modelling bad outcomes as a separate binary risk target;
3. add calibration checks for predicted probabilities;
4. evaluate models on a time-based split, using earlier loans for training and later loans for testing;
5. extend the framework to survival or competing-risks models to use censored active loans directly.

Table 6: Top SHAP features for *Sold*, 60-day random forest.

Feature	Mean abs. SHAP	Mean SHAP
`customer_category_id`	0.0497	0.0497
`w60d_payment_total`	0.0117	0.0095
`w60d_payment_gap`	0.0110	0.0067
`w60d_total_paid_to_due_ratio`	0.0099	0.0084
`w60d_principal_paid`	0.0085	0.0078
`w60d_principal_recovery_rate`	0.0081	0.0051

Table 7: Top SHAP features for *Written Off*, 60-day random forest.

Feature	Mean abs. SHAP	Mean SHAP
`customer_category_id`	0.0432	0.0432
`w60d_days_to_first_payment`	0.0038	-0.0037
`w60d_late_fee_paid`	0.0035	-0.0035
`w60d_payment_gap`	0.0035	-0.0035
`contract_charge_per_day`	0.0027	-0.0007
`w60d_interest_paid`	0.0023	0.0017